

ICT ACADEMY OF KERALA

Certified Specialist in Data Science and Analytics

Assignment on Statistics

Do the following using this dataset ([click here](#)) in google sheet:

1. Calculate the **mean, median, and mode** for the math, reading, and writing scores.
2. Determine the **range, variance, and standard deviation** for each score category.
3. Identify the **mean, median, and mode** of math scores for students with different parental education levels.
4. Compare the **mean scores** of students who took the test preparation course versus those who didn't.
5. Calculate the **interquartile range (IQR)** for each score category.
6. Determine the **standard deviation** for math scores by lunch type (free/reduced vs. standard).
7. Find the **correlation coefficient** between math and reading scores, and then between reading and writing scores.
8. Interpret the strength and direction of the correlation. What might this suggest about the relationship between different scores?
9. Calculate the **covariance** between math and reading scores. How does it compare with the correlation? What does it suggest?
10. **Z-Test:** Assume the national average math score is 50. Use a one-sample z-test to test if this sample's average math score is significantly different from the national average.
11. **T-Test:** Compare the reading scores between students with different lunch types (free/reduced vs. standard) to test if there is a significant difference.
12. **F-Test:** Check if the variances of math scores differ significantly between students with and without test preparation.